

Exploring the Design Space of Headphones as Wearable Public Displays

Dennis Stanke

Pia Brandt

Michael Rohs

{dennis.stanke,michael.rohs}@hci.uni-hannover.de

Leibniz University Hannover

Hannover, Germany

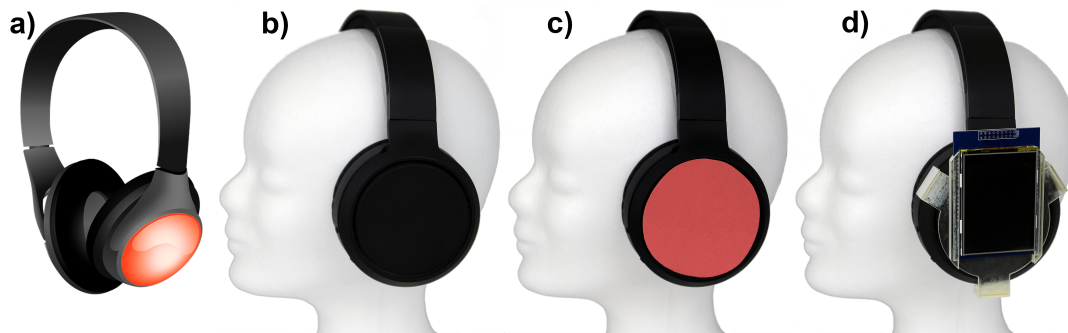


Figure 1: During our design process, we discussed that the ear cup is suited best for showing information to people nearby (a). We decided to use over-ear headphones (b) due to the size advantage. After creating a rough paper prototype (c), we created a 3D-printed clip that holds the wearable public displays to the ear cups (d).

ABSTRACT

The need for online meetings increased drastically during the COVID-19 pandemic. Wearing headphones for this purpose makes it difficult to know when a headphone wearing person is available or in a meeting. In this work, we explore the design possibilities of headphones as wearable public displays to show the current status or additional information of the wearer to people nearby. After two brainstorming sessions and specifying the design considerations, we conducted an online survey with 63 participants to collect opinions of potential users. Besides the preference of the colors red and green as well as using text to indicate availability, we found that only 54 % of our participants would actually wear headphones with public displays attached. The benefit of seeing the current availability status of a headphone-wearing person in an online meeting or phone call scenario were nonetheless mentioned even by participants that would not use such headphones.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices**; *Interaction design*; *Interaction devices*.

KEYWORDS

public displays, wearable devices, headphones, design, acceptability

ACM Reference Format:

Dennis Stanke, Pia Brandt, and Michael Rohs. 2022. Exploring the Design Space of Headphones as Wearable Public Displays. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519756>

1 INTRODUCTION

During the global COVID-19 pandemic, the usage of online meetings and workshops instead of meeting in person increased [18]. In households where several people live and have meetings, the need for headphones is inevitable. This creates the problem that it is unknown whether a person wearing headphones is available or in a meeting and cannot be disturbed. To address this problem we propose to use headphones as a wearable public display. Wearers of such headphones can show their current availability status to people nearby and display additional information, such as social media activities or information about music that they currently listen to. To investigate this idea we analyzed the design space of headphones as wearable public displays with a focus on suitability and acceptance.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519756>

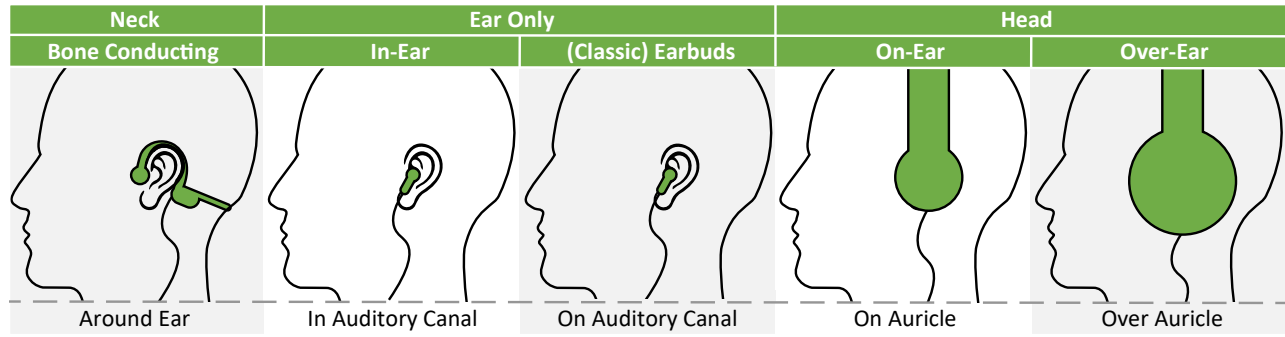


Figure 2: An overview of the headphone types we considered during the brainstorming sessions.

2 RELATED WORK

The majority of research in the area of public displays focuses on large stationary screens [2, 15]. Using smaller public displays as a wearable in a mobile scenario, where the wearer can show information to people nearby, is less explored. One of the first research projects on this topic used classic monochrome LCD screens to show a short text [4]. Cased in a badge or brooch, it can show movie quotes or other information that the wearer is willing to share [5, 7]. Ljungstrand et al. [14] broke the limitation of text and modified a Nintendo Game Boy Color to show colored images worn as a brooch. Furthermore, Colley et al. [6] combined the usage of textual and graphical information to show wellness data – like for how long a person has been sitting on a chair – on a brooch and on a display mounted on a hat. They found that sharing such data to people nearby motivates a healthier lifestyle.

Häkikilä et al. [10] designed a necklace that shows the current busyness status of the wearer to people nearby by showing one of the colors red, yellow or green. They evaluated their wearable in a workplace scenario with 18 participants. Their results show that the necklace increased the participants’ self-awareness concerning busyness and stress level. Further, they showed that people nearby adapt their behavior based on the current busyness status of the wearer.

Alt et al. [1] proposed the idea to show contextual advertising on a public display worn on a person’s back. They used a tablet PC and a backpack with a window. Depending on the user’s location the display shows advertisements of local shops, metro stations, and tourist maps. The approach of a portable billboard was also explored by Kleinman et al. [12]. They used the back of a two-screen laptop to show messages to people nearby. It attracted the attention of others and led to conversations. However, the conversations were mainly about the device itself and not about the message on the device.

The idea of using the smartwatch display as a wearable public display was proposed by Pearson et al. [16]. Besides exploring the design space in multiple studies and naming design considerations, they showed that the concept is socially acceptable. Klamka et al. [11] analyzed the physical design space of smartwatches and straps. They extended the smartwatch display to the watch strap with two additional displays, which can also be used as public displays. To

give other researchers access to their work, they provide an open source research platform.

Groesch [9] patented the idea of a removable media player included in the ear cup of over-ear or on-ear headphones (see Figure 2). The idea to show information on a display that is included in the headband of a headphone has been presented and patented [3, 13] as well. A patent combining both – a display included in the headband and a removable media player with a display attached to the ear cups – was presented by Pong and Liu [17].

3 DESIGN PROCESS

To analyze the design space of headphones as wearable public display, we conducted two brainstorming sessions to develop ideas and to narrow down design choices. The main focus of the brainstorming sessions was to find suitable solutions to show people nearby information on the current status of the headphone’s wearer. This includes the choice of the output modality as well as the input and possible scenarios where such a device could enrich the interaction between two persons. The first brainstorming session was conducted by four students with a technical background from our university. For the second brainstorming session we recruited three additional participants without technical background to discuss the ideas of the first brainstorming session and to gather additional ideas.

The brainstorming sessions showed that head-worn headphones (see Figure 2) are the most promising design to show information to people nearby, especially over-ear headphones. The surface and size of the ear cups allow the addition of a graphical display, which can be seen even from a distance. The headband could also be used as a display space, but information on the band on top of the head is not visible to a person nearby. Other headphones that are worn directly in the ear are less suitable, since the small surface only allows the placement of a single LED and not a spatially extended graphical display. The information shown by this LED could for example be mistaken for the current battery status instead of the status of the person wearing the headphones.

To show information to nearby people, different types of displays were discussed. Electronic paper displays have the benefit of a low power consumption but the disadvantage of a reduced range of colors (typically white, black and red/yellow) and usually no back-light. This means that they are capable of showing a rudimentary

representation of certain information, but more expressive information with colors or images are not possible. Thus electronic paper displays are not suitable for our purpose. Organic light emitting diode (OLED) displays or classic liquid crystal displays (LCD) are more suitable, but both have limitations. New LCD panel technologies, like vertical alignment (VA) and in-plane switching (IPS), have a wider viewing angle and better contrast than classic LCDs but come with a higher power consumption. OLED displays on the other hand have narrower viewing angles but better contrasts and a lower power consumption than LCDs [8].

The participants of our brainstorming sessions found the usage of color coding useful, especially the use of the traffic light colors red, yellow, and green. These colors can indicate the current status of the wearer. Red means that a person is currently not available for conversations, yellow means that the person is busy but interruptible, and green that the person is available for conversations. Within an office environment other colors could indicate specific situations in that company. For example, the color blue could mean that the person is in a meeting or phone call. The addition of pictograms or images could help to show more information about the context of the wearer. Music notes with the addition of text could indicate that the person is currently listening to a specific song. Other symbols like plus, minus, cross, check mark or smiley faces or showing plain text could also help. Showing a QR code, for example as train ticket, is also possible.

In our brainstorming sessions, two main categories of usage scenarios were defined: private and public scenarios. Private scenarios are scenarios in which the wearer of the headphones is in a familiar environment with people known to the wearer. This includes locations like the wearer's home, but also locations like the office of the wearer, where the surrounding is familiar and the people nearby the wearer are already known. Public scenarios on the other hand include locations like public transport or a shopping mall where the surrounding people are unknown to the wearer.

As a potential input method, our participants discussed the usage of a touch screen on the headphones, using the smartphone to control the display. Further they discussed the usage of an automatic detection system that recognizes what the wearer is currently doing.

4 SURVEY

To get an overview of the acceptance of our design considerations we conducted an online survey. Besides querying their headphone usage habits we asked the participants how they would react in specific scenarios containing headphones with displays attached.

4.1 Participants

For our online survey we recruited 63 participants (19 female, 44 male, age 18-71, $\bar{x} = 33.18$, $\sigma = 17.19$). Except for one participant all participants own headphones. They use over-ear (40 participants), in-ear (33 participants), classic earbuds (20 participants), and on-ear (18 participants) headphones with a microphone (49 participants). Bluetooth (39 participants), noise cancellation (24 participants), illumination via LEDs (9 participants), and touch controls (1 participant) were used as additional features. The participants use their headphones at home (40 participants), at work (39 participants), on the train (35 participants), or generally when they are on the

move (33 participants) and while pursuing activities like relaxing (38 participants), playing games (37 participants), during sport (26 participants), shopping (17 participants), or playing a keyboard instrument (1 participant). The headphones are used by our participants to listen to content like music or podcasts (60 participants), make phone calls (26 participants), suppress surrounding noises (24 participants), avoid being distracted (15 participants), and as a social signal not to be disturbed (5 participants).

4.2 Results

We explained a travel scenario to our participants where the participant wants to talk to another passenger who is wearing headphones. 42 participants would speak directly to the person if the matter was urgent. 10 participants would try to get the attention of the passenger before speaking. 7 participants would speak directly to the passenger and 4 participants would not talk to the passenger in such a situation. We also asked the participants whether the type of headphones have an influence on their decision in this scenario. Only 8 participants would change their behavior based on the headphone type. They explained this by stating that they would not talk to passengers with bigger sized headphones (5 participants). The possibility to miss smaller headphones and thus speak directly to the passenger was also mentioned (3 participants).



Figure 3: We showed our participants this sketch of headphones with light emitting displays on the ear cups and asked them about the meaning of the colors red and green.

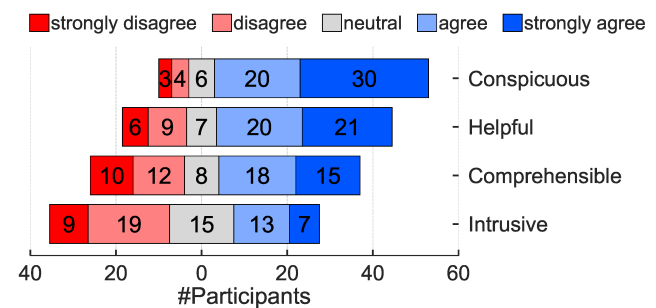


Figure 4: According to our participants, the use of colors as indicator for the current availability status is conspicuous, helpful, and comprehensible. They have a mixed opinions on the level of intrusiveness of the colors.



Figure 5: We presented our participants different icons (first number; e.g. 1.x) as well as text to express the current availability status of the wearer. Besides a ranking of the different icons, the participants were asked whether they would prefer a colored, white or black background (second number; e.g. x.1).

We asked the participants about their opinion of the meaning of the colors red and green on LED supported headphones (see Figure 3). Multiple answers were possible. The majority of the participants answered that it is a decorative element of the headphones without any function (33 participants). 17 participants said that the color could indicate whether the wearer of the headphones wants to be disturbed. Some of the participants also mentioned that it could show the current status of the system (like on or off; 7 participants), the battery status (6 participants), the status of the user (like mood; 4 participants), or whether a noise cancellation function is activated (2 participants). When the headphones would glow green or red, 42 participants would not change their behavior of speaking to the wearer. The other 21 participants stated that they would rather speak to the person when the headphones glows green and would rather not speak to the person when it glows red.

Subsequently, we told the participants that the colors green and red could indicate whether the wearer can be spoken to. As shown in Figure 4, the participants judged the usage of colors to show the current availability status of the wearer conspicuous (50 participants), helpful (41 participants), and comprehensible (33 participants). They did not have a differing opinions regarding intrusiveness. 16 participants thought that additional colors, like yellow and blue, could be helpful to show a specific status.

The second presented scenario was an office scenario where the participant wants to talk to a colleague who wears headphones. In an urgent matter, 33 of the participants would directly speak to the colleague. 14 participants would try to get the attention before speaking. 11 participants would directly speak to the colleague, even if the matter is not urgent. 4 participants would not speak to the colleague and 1 participant would write an email instead of speaking directly to the colleague who wears headphones. Additionally, we asked our participants whether a corporate wide-color that indicates special situations could be helpful. We therefore defined



Figure 6: We showed our participants examples of how the shared information could be visualized on the headphones.

that blue means that the colleague is in a meeting. 57 participants thought that such a defined meaning of a color would help them to make a decision about whether or not to talk to the colleague. The other 6 participants thought that this is not helpful. They justified this by stating that color codes are not necessary (1 participant), the wearer could have forgotten to change the color back (1 participant), they would not talk to people with headphones anyway (1 participant), and that the colors red and green are better suited for the purpose (1 participant). We asked the participants whether they could imagine icons as additional information to the color shown on the headphones. 31 participants thought that additional icons are useful and 32 participants thought that colors are sufficient.

Based on the collected ideas of the brainstorming sessions, we created a set of icons, which could be shown on the headphones (see Figure 5). We asked the participants to rank them by preference. As seen in Table 1, our participants prefer colors and simple text to show their current status to people nearby.

Besides the ranking of the icons we also asked the participants whether they would prefer a colored background (like red and

Icon (<i>available</i> ; Figure 5 left)	M	$\bar{x}$	Icon (<i>not available</i> ; Figure 5 right)	M	$\bar{x}$
Green color (6.x)	2.0	3.19	Text <i>do not disturb</i> (1.x)	3.0	3.52
Text <i>available</i> (1.x)	2.0	3.21	Red color (6.x)	3.0	3.62
Check mark (4.x)	3.0	4.21	X (4.x)	4.0	4.25
Happy face (2.x)	4.0	4.19	Crossed out speech bubble (5.x)	5.0	4.44
Speech bubble (5.x)	5.0	4.75	Minus sign (3.x)	5.0	4.79
Plus sign (3.x)	7.0	5.86	Sad face (2.x)	6.0	5.37

Table 1: The participants ranked the 6 icons according to their preferences. They only had to rank the symbol they could imagine using to display the status *available* (left table) and *not available* (right table), respectively. If a symbol was not ranked by a participant, we gave the symbol a rank of 7. The two right hand columns of each table show median (M) and mean rank ($\bar{x}$).

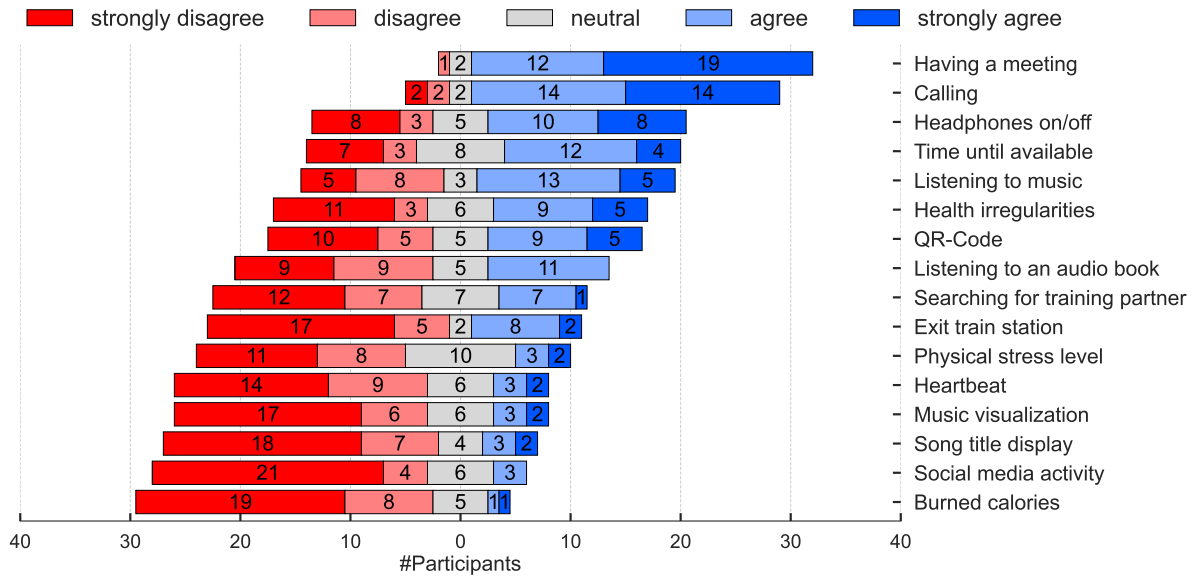


Figure 7: We asked our participants which information they would share while wearing headphones with public displays. The majority of the 34 participants that could imagine wearing such headphones would share the information that they are currently in a meeting or call as well as the information whether their headphones are on or off and that they are listening to music. Sharing more sensitive information, like burned calories or social media activity, were generally rejected by our participants.

green), a transparent or black background, or a white background for the icons. The majority of the participants preferred a colored background (35 participants) over a transparent or black background (19 participants), or a white background (9 participants). For the situation that the wearer of a headphone with displays wants to show multiple information items on the displays we asked our participants which of the following they would find useful: display multiple icons at the same time, display multiple icons alternately, and display additional text (multiple answers were possible). 38 participants stated that it is useful to display several icons alternately, 19 participants found the display of multiple icons at the same time useful, and 17 participants judged the display of additional text useful. 5 participants think that none of these are useful (without naming alternatives).

We asked our participants which information they would imagine sharing while using such headphones and showed them examples (see Figure 6). Our participants would share the information that they are currently in a meeting (42 participants) or phone call (37 participants). Participants who could imagine wearing such headphones were also willing to share whether their headphones are on or off and while they are listening to music (see Figure 7).

As a final question, we asked our participants whether they could imagine using headphones with wearable public displays to show their current status. 29 of the 63 participants would not wear such headphones in public. They argued that they are only wearing small in-ear headphones (3 participants) and that they have privacy concerns (3 participants). 5 participants would not wear such headphones privately, but could imagine wearing them

at work. However, 34 participants would use headphones with displays that show their current status.

5 PROTOTYPE

With the collected ideas and insights of possible users, we created an online available prototype¹. We decided to use Philips TAH5205 over-ear headphones due to the size advantage of the ear cups (see Figure 1b). The prototype itself consists of two parts: an ESP32 with two displays and an Android application. To attach the displays on the ear cups, we created a removable custom 3D-printed clip (see Figure 1d). The 2.8-inch resistive touch screens have a resolution of 320x240 pixels and use the ILI9341 driver chip. The displays are directly wired to the ESP32, which communicates with an Android phone via Bluetooth. An Android application controls the contents of the screens which is a combination of colors, text, and icons. Besides the support of the icons we created (see Figure 5), the application also supports showing specific application icons as exemplified in Figure 6 to show the current status to people nearby.

6 DISCUSSION

The results of our online survey show that 54 % of our participants would use headphones with wearable public displays that show their current status. Although we presented multiple areas of application, our participants think that the most promising use cases for such headphones are while having a meeting (66.7 %) and during a phone call (58.7 %). Even participants who would privately not use such headphones see a benefit in these two scenarios. The other scenarios were generally not accepted by the majority of our participants. However, participants who see a benefit in display-supported headphones could also imagine showing whether the headphones are turned off (52.9 %) and showing that they are listening to music (52.9 %; see Figure 7). Showing additional information on the headphones – like the artist or title of a song or sensitive information like burned calories – were rejected by our participants.

In one of our first questions, we asked our participants about the meaning of the colors red and green on a headphone sketch (see Figure 3). Interestingly, 33 of our participants thought that the colors are a decorative element, while later only 24 participants stated that the color is not comprehensible as the current availability status (see Figure 4). Further, the majority of our participants ranked the colors as the most useful element for showing the current availability status of the wearer (see Table 1). This shows that color alone is not sufficient to understand the meaning and at least additional text is necessary.

One limitation of our work is that we asked our participants only about their opinion and acceptance as a wearer of the headphones and not as someone who receives information from another person's headphones. Having analyzed the online survey, we think that using wearable public displays on headphones as advertising or news spaces could be an interesting use case that should be further explored. Another limitation is that the results are based on an online survey only, instead of an in-the-wild study followed by interviews. The latter gets bystanders to react immediately and intuitively when seeing the headphones, rather than having the time to think about how they would react, like was the case in the survey.

¹<https://www2.hci.uni-hannover.de/research/headphone-display>

Furthermore, we have not focused our exploration on a specific group of people. We now think that groups like indie musicians or social media influencer may benefit from headphones as wearable public displays, as they could reach an additional audience. This should also be explored in the future.

7 CONCLUSION

We proposed the usage of headphones as wearable public displays to show the current status of the wearer or to share other information to people nearby. Furthermore, we drew several insights from two brainstorming sessions, the resulting design process, an online survey with 63 participants, and from developing our prototype. We showed the suitability of headphones as wearable public displays, with limitations regarding acceptability. The feedback of our participants shows that display-equipped headphones are acceptable particularly in non-public scenarios. Such scenarios include working in the office or at home with other people nearby, when regularly participating in online meetings. Showing the current availability status of the wearer in such a situation, was considered as beneficial by our participants. In public scenarios on the other hand, display-equipped headphones were regarded as less acceptable according to our participants.

ACKNOWLEDGMENTS

We would like to thank Kevin Schumann, Daniel Wittenburg, Michael Mircea, Nico Ostendorf and Nikita Povelkin for their contribution to the creation of the prototype.

REFERENCES

- [1] Florian Alt, Albrecht Schmidt, and Christoph Evers. 2009. Mobile contextual display systems. *Adj. Proc. of HotMobile* 9 (2009).
- [2] Florian Alt, Stefan Schneegaß, Albrecht Schmidt, Jörg Müller, and Nemanja Memorovic. 2012. How to Evaluate Public Displays. In *Proceedings of the 2012 International Symposium on Pervasive Displays* (Porto, Portugal) (*PerDis '12*). Association for Computing Machinery, New York, NY, USA, Article 17, 6 pages. <https://doi.org/10.1145/2307798.2307815>
- [3] Michael Boateng and George Nyamie. 2009. Headphone display. US Patent App. 29/318,834.
- [4] Richard Borovoy, Fred Martin, Sunil Vemuri, Mitchel Resnick, Brian Silverman, and Chris Hancock. 1998. Meme Tags and Community Mirrors: Moving from Conferences to Collaboration. In *Proceedings of the 1998 ACM Conference on Computer Supported Cooperative Work* (Seattle, Washington, USA) (*CSCW '98*). Association for Computing Machinery, New York, NY, USA, 159–168. <https://doi.org/10.1145/289444.289490>
- [5] Richard Borovoy, Michelle McDonald, Fred Martin, and Mitchel Resnick. 1996. Things that blink: Computationally augmented name tags. *IBM Systems Journal* 35, 3.4 (1996), 488–495.
- [6] Ashley Colley, Bastian Pfleging, Florian Alt, and Jonna Häkklä. 2020. Exploring public wearable display of wellness tracker data. *International Journal of Human-Computer Studies* 138 (2020), 102408. <https://doi.org/10.1016/j.ijhcs.2020.102408>
- [7] Jennica Falk and Staffan Björk. 1999. The BubbleBadge: A Wearable Public Display. In *CHI '99 Extended Abstracts on Human Factors in Computing Systems* (Pittsburgh, Pennsylvania) (*CHI EA '99*). Association for Computing Machinery, New York, NY, USA, 318–319. <https://doi.org/10.1145/632716.632909>
- [8] María Rodríguez Fernández, Eduardo Zalama Casanova, and Ignacio González Alonso. 2015. Review of Display Technologies Focusing on Power Consumption. *Sustainability* 7, 8 (2015), 10854–10875. <https://doi.org/10.3390/su70810854>
- [9] Michael R Groesch. 2012. Headphones with embeddable accessories including a personal media player. US Patent 8,213,666.
- [10] Jonna Häkklä, Romina Poguntke, Emmi Harjuniemi, Lauri Hakala, Ashley Colley, and Albrecht Schmidt. 2020. *BuSiNec - Studying the Effects of a Busyness Signifying Necklace in the Wild*. Association for Computing Machinery, New York, NY, USA, 2177–2188. <https://doi.org/10.1145/3357236.3395455>
- [11] Konstantin Klamka, Tom Horak, and Raimund Dachselt. 2020. *Watch+Strap: Extending Smartwatches with Interactive Strap Displays*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3313831.3376199>

- [12] Lisa Kleinman, Tad Hirsch, and Matt Yurdana. 2015. Exploring Mobile Devices as Personal Public Displays. In *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services* (Copenhagen, Denmark) (*MobileHCI '15*). Association for Computing Machinery, New York, NY, USA, 233–243. <https://doi.org/10.1145/2785830.2785833>
- [13] Noel Lee, Jacky Hsiung, and Kendrew Lee. 2014. Headphones with interchangeable decor strip. US Patent 8,861,767.
- [14] P. Ljungstrand, S. Bjork, and J. Falk. 1999. The WearBoy: a platform for low-cost public wearable devices. In *Digest of Papers. Third International Symposium on Wearable Computers*. IEEE, New York, NY, USA, 195–196. <https://doi.org/10.1109/ISWC.1999.806926>
- [15] Jörg Müller, Florian Alt, Daniel Michelis, and Albrecht Schmidt. 2010. Requirements and Design Space for Interactive Public Displays. In *Proceedings of the 18th ACM International Conference on Multimedia* (Firenze, Italy) (*MM '10*). Association for Computing Machinery, New York, NY, USA, 1285–1294. <https://doi.org/10.1145/1873951.1874203>
- [16] Jennifer Pearson, Simon Robinson, and Matt Jones. 2015. It's About Time: Smartwatches as Public Displays. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (*CHI '15*). Association for Computing Machinery, New York, NY, USA, 1257–1266. <https://doi.org/10.1145/2702123.2702247>
- [17] Ronald Pong and King Hei Liu. 2016. Headphones with interactive display. US Patent 9,445,172.
- [18] Kate Puddister and Tamara A. Small. 2020. Trial by Zoom? The Response to COVID-19 by Canada's Courts. *Canadian Journal of Political Science* 53, 2 (2020), 373–377. <https://doi.org/10.1017/S0008423920000505>